

RESEARCH INTERESTS

Formal Methods
Verification
Archimedean Quadratic Modules
Gröebner basis algorithms
Quantifier-free interpolation algorithms for decidable logics
Non-classical logics

EDUCATION

University of New Mexico Albuquerque, New Mexico
Ph.D. in Computer Science, Advisor: Prof. Deepak Kapur 2020–2026
– Thesis: Computing Certificates of Members in Archimedean Quadratic Modules in $\mathbb{A}[X]$ and Certifying the Emptiness in Inconsistent Monogenic Archimedean Quadratic Modules in $\mathbb{A}[\overline{X}]$

University of New Mexico Albuquerque, New Mexico
M.S. in Computer Science, Advisor: Prof. Deepak Kapur 2016–2020
– Thesis: Implementation of Uniform Interpolation Algorithms

Universidad de las Americas Puebla Cholula, Puebla
B.S. in Electronics Engineering, Advisor: Prof. Mauricio Javier Osorio Galindo 2010–2015
– Thesis: Revisiting C_1

RESEARCH EXPERIENCE

University of New Mexico Albuquerque, New Mexico
Research Assistant; Advisor: Prof. Deepak Kapur Fall 2020 - Spring 2026
– Research on Formal Methods and Verification
– Assisted with research on symbolic computation and its application to program analysis.
– Research on computational real algebraic geometry with focus on computing certificates for elements in Archimedean Quadratic Modules.

Microsoft Research Redmond, Washington
Research Intern; Mentor: Principal RSDE Mark Marron Summer 2019
– Verification in Bosque
– Developed a prototype of the verification engine for the Bosque programming language in F^* . Bosque is a language that does not implement loops but offers to programmers transformers and functional programming constructions (limited fold operation) to do their programming tasks.

Universidad de las Americas Puebla Cholula, Puebla
Research Student; Advisor: Prof. Mauricio J. Osorio Galindo 2015-2017
– Research on Paraconsistent Logics
– Collaborated with a group of researchers on Paraconsistent Logics. My activities included working on some theorems and generate models using the answer set solver Clasp.

Universidad de las Americas Puebla Cholula, Puebla
Intern; Advisor: Prof. Ofelia Cervantes Gutierrez Summer 2015
– Innova4D

- Analyzed and implemented graph algorithms to compute Freeman centralities for the development of a recommendation system.

PUBLICATIONS

- [1] **J. Castellanos Joo** and D. Kapur, *Computing certificates in archimedean univariate saturated quadratic modules*, 2026. arXiv: [2605.18980](#) [[cs.SC](#)].
- [2] W. Shang, **J. Castellanos Joo**, C. Mou, and D. Kapur, *Computing certificates of strictly positive polynomials in archimedean quadratic modules*, 2025. arXiv: [2503.11119](#) [[math.AC](#)].
- [3] **J. Castellanos Joo**, S. Ghilardi, A. Gianola, and D. Kapur, “AXDInterpolator: A tool for computing interpolants for arrays with maxdiff”, in *19th International Workshop on Satisfiability Modulo Theories co-located with 33rd International Conference on Computer Aided Verification (CAV 2021)*, CEUR-WS.org, vol. 2908, 2021, pp. 40–52.
- [4] M. Osorio and **J. Castellanos Joo**, “Equivalence among RC -type paraconsistent logics”, *Logic Journal of IGPL*, jzw065, Jan. 2017, ISSN: 1368-9894. DOI: [10.1093/jigpal/jzw065](#)
- [5] M. Osorio, J. L. Carballido, C. Zepeda, and **J. Castellanos Joo**, “Weakening and extending $\mathbb{Z}$ ”, *Logica Universalis*, vol. 9, no. 3, pp. 383–409, Aug. 2015, ISSN: 1661-8300. DOI: [10.1007/s11787-015-0128-6](#)
- [6] M. Osorio and **J. Castellanos Joo**, “A single proof of classical behaviour in da Costa’s C_n systems”, *Electronic Notes in Theoretical Computer Science*, vol. 315, pp. 3–16, Sep. 2015, ISSN: 1571-0661. DOI: [10.1016/j.entcs.2015.06.002](#)

TALKS

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| Computing Certificates of Members in Archimedean Quadratic Modules in $\mathbb{A}[X]$ and Certifying the Emptiness in Inconsistent Monogenic Archimedean Quadratic Modules in $\mathbb{A}[\overline{X}]$
Thesis Defense, University of New Mexico | September, 2026 |
| Computing certificates in compact quadratic modules in $\mathbb{R}[x]$
Proposal Thesis Defense, University of New Mexico | September, 2023 |
| AXDInterpolator: A Tool for Computing Interpolants for Arrays with MaxDiff
19th International Workshop on Satisfiability Modulo Theories. | July, 2021 |
| Implementation of Uniform Interpolation Algorithms
Master Thesis Defense, University of New Mexico | October, 2020 |
| A new interpolation algorithm for the theory of Equality with Uninterpreted Functions
Computer Science Colloquium Series, University of New Mexico | September, 2020 |
| A Single Proof of Classical Behaviour in da Costa’s C_n systems
Ninth Latin American Workshop on Logic/Languages, Algorithms and New Methods of Reasoning LANMR | November, 2014 |

TEACHING ASSISTANT EXPERIENCE

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| Grader at University of New Mexico
<i>CS 561 - Algorithms and Data Structures with Prof. Jared Saia</i> | Fall 2023 |
| Teaching Assistant at University of New Mexico
<i>CS 105L - Introduction to Computer Programming with Prof. Soraya Abad-Mota</i> | Fall 2023 |
| Teaching Assistant at University of New Mexico
<i>CS 357 - Declarative Programming with Prof. Lance Williams</i> | Spring 2023 |
| Head Teaching Assistant at University of New Mexico
<i>CS 241 - Data Organization using C with Prof. Soraya Abad-Mota</i> | Fall 2022 |
| Teaching Assistant at University of New Mexico
<i>CS 429/529 - Machine Learning with Prof. Trilce Estrada</i> | Spring 2022 |

Teaching Assistant at University of New Mexico <i>CS 530 - Geometric and Probabilistic Methods in Computer Science with Prof. Lance Williams</i>	Fall 2019
Teaching Assistant at University of New Mexico <i>CS 500 - Theory of Computation with Prof. Deepak Kapur</i>	Spring 2019
Teaching Assistant at University of New Mexico <i>CS 561 - Algorithms and Data Structures with Prof. Jared Saia</i>	Fall 2018

MENTORSHIP

Brandon Hidalgo (Bachelor) Computer Science department, University of New Mexico <i>Interface construction for a singular package that implements the algorithm in https://dl.acm.org/doi/10.1145/1993886.1993918. I provided support about documentation of Singular and the student built a Jupyter notebook interface in Java to handle input/output for the system.</i>	Fall 2024 - Spring 2025
Abigail Pribisova (Bachelor) Computer Science department, University of New Mexico <i>Implementation of an algorithm for the theory of contiguous arrays equipped with a max diff operator . The deliverables of this project were a poster presented by the student Abigail at the 18th Annual Computer Science Student Conference 2023 at UNM and a working prototype of the interpolation algorithm.</i>	Fall 2022 - Spring 2023

SKILLS

- Programming languages
 - Imperative: C/C++, Java, Go, Julia, Rust
 - Scripting: Python, Bash, Makefile
 - Functional: Haskell, Ocaml, Scala
 - Verification: Z3, Mathsat, SMTInterpol, F^* , Prover9, Mace4, Lean
 - Symbolic/Algebraic: Mathematica, Maple, Macaulay2, Singular
 - Document typesetting: $\text{\LaTeX}$, Pandoc, Madoko, Markdown, Org
 - Web design: HTML, CSS, Javascript, Typescript, Hugo

LANGUAGES

- **English:** Fluent
- **Spanish:** Native

SOFTWARE PROJECTS

SatQMCert	2026
<i>A maple package to compute certificates of members in Archimedean Saturated Quadratic Modules. The algorithm is fully symbolic in the sense that it does not depend on numerical algorithm to obtain certificates.</i>	
BasicLemma	2026
<i>A maple package to compute Bezout-like polynomials for coprime and non-negative polynomials over compact semialgebraic sets. The package BasicLemma implements the algorithm in Lemma 2.1 from Positivity, sums of squares and the multi-dimensional moment problem II.</i>	
StrictlyPositiveCert	2026
<i>A maple package to compute certificates of strictly positive polynomial over Archimedean Quadratic Modules. The package StrictlyPositiveCert implements Algorithm 4 in [2], which computes certificates of strictly positive polynomials over basic semialgebraic sets associated a set of generators for which the quadratic module generated by is Archimedean in the multivariate case.</i>	
haskell-grader	2023
<i>Haskell application to automatically evaluate student assignments for CS 357: Declarative Programming at UNM.</i>	
AXDInterpolator	2021
<i>This project implements an interpolation algorithm proposed in FoSSaCS 2021 using the Z3 API. The project allows the user to choose Z3, Mathsat, or SMTInterpol as interpolation engines. The tool returns a formula in SMTLIB2 format, which allows compatibility with model checkers and invariant generators using such a format.</i>	

EUFInterpolator	2020
<i>Master thesis work implementing new interpolation algorithms for the theory of equality and uninterpreted functions (EUF), octagonal formulas, and its combination.</i>	
Bosque Transpiler to F*	2019
<i>Prototypical implementation of a transpiler embedding a subset of the Bosque semantics into the Proof-oriented programming language F*.</i>	

WORKSHOPS ATTENDED

Satisfiability: Theory, Practice, and Beyond <i>Beyond Satisfiability</i>	2021
Satisfiability: Theory, Practice, and Beyond <i>Theoretical Foundations of SAT/SMT Solving</i>	2021
AMS Short Course <i>Sum of Squares: Theory and Applications</i>	2019

CONFERENCE REFEREEING

Thirteen Latin America Workshop on New Methods of Reasoning <i>Reviewer</i>	2020
35th International Conference on Logic Programming <i>Reviewer</i>	2019
11th Latin American Workshop on New Methods of Reasoning <i>PC member</i>	2018
14th Annual Computer Science Student Conference <i>Reviewer</i>	2018
17th Latin American Symposium on Mathematical Logic <i>Reviewer</i>	2017
10th Latin American Workshop on Logic/Languages, Algorithms and New Methods of Reasoning <i>Reviewer</i>	2016
8th Mexican Congress on Artificial Intelligence <i>Reviewer</i>	2016
12th International Colloquium on Theoretical Aspects of Computing <i>Reviewer</i>	2015

SCHOLARSHIPS AND AWARDS

Travel Scholarship for OPLSS <i>Travel scholarship to attend Oregon Programming Languages Summer School</i>	2017
AMIGO Scholarship <i>Scholarship for Graduate Studies at the University of New Mexico</i>	2016 - 2018
ANFEI <i>Best student of the Electronics Engineering 2015 class</i>	2015
Magna Cum Laude (BSc) <i>Universidad de las Americas Puebla.</i>	2015
Roberto Rocca Scholarship <i>Scholarship for Undergraduate Studies at Universidad de las Americas Puebla</i>	2014

SOCIETY MEMBERSHIPS

Women in Computing association at the University of New Mexico.

SERVICE

CS Advisory Board

University of New Mexico

Graduate Student Representative

2021 - 2023

- Participated in discussions about the state of the department and proposal of new initiatives. regarding graduate and undergraduate matters, as well as research and the position of the department within the university.

CS Graduate Student Association

University of New Mexico

Treasurer

2017 - 2018

- Developed website for the Computer Science Student Conference 2018 at UNM and keep track of Internal Requisitions.

Clique Student Organization

Universidad de las Américas Puebla

Founder Member

2014 - 2015

- This organization provided students a proper environment to develop programming skills for programming competitions like the ACM ICPC.